



Retrieval Contrastive Learning for Aspect-Level Sentiment Classification

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ABSTRACT

Aspect-Level Sentiment Classification (ALSC) aims to assign specific sentiments to a sentence toward different aspects, which is one of the crucial challenges in the field of Natural Language Processing (NLP). Despite numerous approaches being proposed and obtaining prominent results, the majority of them focus on leveraging the relationships between the aspect and opinion words in a single instance while ignoring correlations with other instances, which will make models inevitably become trapped in local optima due to the absence of a global viewpoint. Instance representation derived from a single instance, on the one hand, the contained information is insufficient due to the lack of descriptions from other perspectives; on the other hand, its stored knowledge is redundant since the inability to filter extraneous content. To obtain a polished instance representation, we developed a Retrieval Contrastive Learning (RCL) framework to subtly extract intrinsic knowledge across instances. RCL consists of two modules: (a) obtaining retrieval instances by sparse retriever and dense retriever, and (b) extracting and learning the knowledge of the retrieval instances by using Contrastive Learning (CL). To demonstrate the superiority of RCL, five ALSC models are employed to conduct comprehensive experiments on three widely-known benchmarks. Compared with the baselines, ALSC models achieve substantial improvements when trained with RCL. Especially, ABSA-DeBERTa with RCL obtains new state-of-the-art results, which outperform the advanced methods by 0.92%, 0.23%, and 0.47% in terms of Macro F1 gains on Laptops, Restaurants, and Twitter, respectively.

1. Introduction

As an important technique for real-world scenarios of public opinion monitoring (Chen, Zhang, Xu and Cao, 2022; Li, Wang and Wang, 2021) and content recommendation (Elahi, Kholgh, Kiarostami, Oussalah, & Saghari, 2023), Aspect-Level Sentiment Classification (ALSC) has drawn more attention in recent years. Unlike coarse-grained sentiment analysis, which assigns a sentiment polarity to the entire sentence, ALSC seeks to distinguish the sentiment polarities of a sentence toward different aspects. As illustrated in Fig. 1, the sentiment polarities of aspects “quality”, “performance” and “battery life” in the sentence “Overall I feel this netbook was poor quality, had poor performance, although it did have great battery life when it did work”. are identified as “Negative”, “Negative” and “Positive”, respectively. For the sake of clarity, the following sections will use the term “instance” to refer to a sentence that concerns a certain aspect. Hence, the above sentence contains three instances that each relate to one of three aspects.

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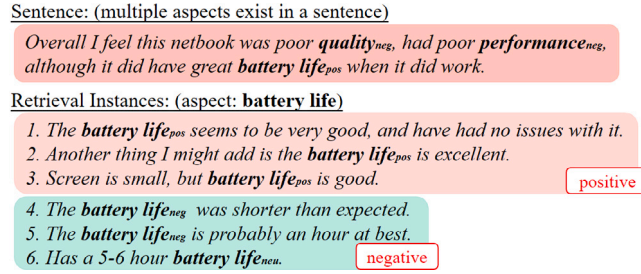


Fig. 1. A sentence with three aspects and their sentiment polarities.

Many ALSC studies have been done over the past decades using a variety of techniques to reveal the genuine interrelation of aspect and context, including rule-based, machine learning-based, and deep learning-based methods (Zhang, Li, Deng, Bing and Lam, 2022). Due to the superiority in learning aspect-related semantic representations, deep learning-based methods have gradually risen to supremacy in recent years (Brauwert & Frasincar, 2022). Hence, we concentrate on the deep learning-based approaches (Birjali, Kasri, & Beni-Hssane, 2021; Joseph, Vineetha, & Sobhana, 2022), which can be divided into RNN-based, Graph-based, PLM-based, and Knowledge-based methods. RNN-based methods (Cheng et al., 2017; Tay, Tuan, & Hui, 2018) concentrate on the superficial sequence information of words while Graph-based approaches (Chen, Teng, Wang and Zhang, 2022; Huang & Carley, 2019; Li, Chen et al., 2021; Phan, Nguyen, & Hwang, 2023; Wang, Shen, Yang, Quan, & Wang, 2020; Zhu, Zhang et al., 2023) aim to reveal the intrinsic relationships. RNN-based methods still suffer from the problem of long-term dependency although have been alleviated by the variants of LSTM and GRU. In addition, introducing the attention mechanism enables the RNN-based methods to review the important part of the sentence and improves the model significantly (Su et al., 2021). However, the accuracy of the important contents determined by the attention mechanism is not guaranteed, especially for long sentences with complex structures. To explicitly ascertain the accurate opinion words, with the help of syntactic parsers like StanfordNLP (Manning et al., 2014), researchers are increasingly using graph-based approaches to encode sentence that takes the dependency relations between words into account. Despite graph-based approaches possessing great interpretability when judging the instance's sentiment, they depend highly on the accuracy of the parser, which is another challenge in the Natural Language Processing (NLP) community.

Apart from custom-designed models from scratch, integrating pre-trained language models (PLMs) such as BERT (Devlin, Chang, Lee, & Toutanova, 2019) on ALSC task have brought substantial improvements in recent years. ALSC models with PLMs serving as their backbone have now taken center stage (Liang, Wei, Mao, Wang, & He, 2022). Since PLMs are trained on sizable amounts of unlabeled corpora, a significant quantity of general knowledge and linguistic grammar is implicitly stored in the model parameters for retrieval by subsequent tasks. Compared with RNN-based and Graph-based methods that only store the training data's knowledge, PLM-based models (Dai, Yan, Sun, Liu, & Qiu, 2021; Li, Bing, Zhang, & Lam, 2019; Yong, Chen, Wang, Xiao, & Tang, 2023) contain the knowledge of pre-trained corpus additionally, and bring marvelous results in many NLP tasks. Known as a black box, the role of PLM is inexplicable when retrieving knowledge from the model parameters. Benefiting from advancements in knowledge graph and retrieval technology, external knowledge is introduced to improve the accuracy of the ALSC model and enhance their interpretability (Jin, Zhao, Zhang, Liu, & Yu, 2023; Wu, Huang, & Deng, 2023; Zhong et al., 2023; Zhou, Huang, Hu and He, 2020), however, this improvement comes at an enormous memory consumption cost.

Most existing ALSC methods are concentrated on modeling a single sentence, where RNN-based and Graph-based methods model the relationships of aspect and context words in a single sentence, and PLM-based and Knowledge-based methods enhance the semantics of each word or phrase. Despite their prominence, we argue that modeling a single sentence does not provide a comprehensive picture of the distribution of the dataset, and these methods will inevitably become trapped in local optima, which results in flawed instance representation and limits the model performance. On the one hand, the useful information provided by a single sentence is limited, it is useful to enhance the semantics of the sentence from different perspectives; on the other hand, the redundant information will inevitably be taken into account when modeling the sentence. To solve the two issues simultaneously and obtain polished instance representations, inspired by REINA (Wang et al., 2022) that retrieve knowledge from training data, we attempt to retrieve aspect-related instances from training data and extract the intrinsic knowledge across instances. As illustrated in Fig. 1, for the target instance with the aspect of "battery life", we first retrieve instances with the same aspects from training data, dubbed retrieval instances. Although these instances are different, they are all descriptions associated with the aspect of "battery life", and can provide information supplement to the target instance. Hence, after obtaining the retrieval instances, we employ Contrastive Learning (CL) to extract intrinsic knowledge across instances. On the one hand, retrieving related information from training data rather than external knowledge can avoid huge storage space and computing costs; on the other hand, extracting the commonality and ignoring the individuality from retrieval instances are conducive to filtering out redundant information in the target instance and generating a more robust instance representation.

To this end, a novel Retrieval Contrastive Learning (RCL) framework is proposed in this paper, RCL is model-agnostic and has high scalability and flexibility, which consists of two modules: (a) retrieving instances from training data, and (b) extracting common features from the retrieval instances. To be more precise, we first index the training instances into a list of {key, value} pairs, where key refers to the aspect, and value refers to the entire instance. And then, instances with the same aspects as the target

instance are retrieved by querying the constructed index, known as the sparse retriever. Furthermore, a dense retriever is utilized to supplement the retrieval instances if they are insufficient. After obtaining the retrieval instances, CL is employed to link the retrieval instances and the target instances, which normalizes the instance representation space and enables models to generate more reliable representations for downstream tasks. To confirm the efficacy of our proposed RCL framework, some exploratory experiments on three popular domain datasets (Restaurants, Laptops, and Twitter) are conducted. The major contributions of our work are:

- To obtain a polished instance representation for ALSC, we proposed an RCL framework to link instances with the same or similar aspects and extract the intrinsic knowledge from these associated instances by using CL. To the best of our knowledge, we are the first to explicitly consider the relationships of instances with the same or similar aspects.
- We index the training instances with Elasticsearch and design a sparse retriever to get the instances with the same aspects, a dense retriever to supplement the retrieval instances if they are in deficiency. On the one hand, retrieval instances are the supplement of the target instance, which can increase the information of instance representation; on the other hand, extracting the features shared by the retrieval instances and the target instance can effectively filter the redundant knowledge in the instance representation.
- We do extensive experiments to verify RCL framework can help the existing ALSC models achieve more elaborate instance representations for sentiment classification. Experimental results on three widely-known benchmarks demonstrate the superiority of our proposed RCL, where ABSA-DeBERTa obtains new state-of-the-art results after being trained with RCL.
- The case study shows that RCL can enable the model to (1) understand the semantics of the unseen aspects; (2) filter out noise in the text; (3) be aware of implicit opinions, and then generate richer semantic representations for sentiment prediction.

Following is the paper's structure of the remainder: some relevant research contents are introduced in Section 2. The methodology of RCL is described in detail in Section 3. Sections 4 and 5 each provide information on the experimental design and outcome analysis. Section 6, offers concluding observations. Finally, we discuss our work in Section 7.

2. Background

2.1. Aspect-level sentiment classification

As an important component of NLP, Sentiment Analysis (SA) intends to find people's sentimental state, opinions, and attitudes to a product or service automatically, which contains coarse-grained SA that assigns a holistic sentiment tendency to the whole sentence roughly and fine-grained SA that identifies the sentiment polarities of sentence toward different aspects. Aspect-Based Sentiment Analysis (ABSA), also known as fine-grained SA, can be divided into single ABSA and compound ABSA tasks (Zhang, Li et al., 2022). And the ALSC task, which seeks to forecast the sentiment polarity for a sentence toward different aspects, is a part of single ABSA tasks. Recently, ALSC has been dominated by neural-based ALSC models, including RNN-based, Graph-based, PLM-based, and Knowledge-based models, which concentrate on modeling the interaction between the aspect and its context to capture the beneficial features to predict the sentiment polarity correctly.

RNN-based ALSC models: Recurrent Neural Network (RNN) is suggested as a specific way to manage variable-length sequence data, and it readily fits the bill for tasks involving sentence modeling like the ALSC challenge. However, vanilla RNN suffers from the problem of long-term dependency. Hence, LSTM, a variant of RNN that alleviates the problem of long-term dependency by the gate mechanism, is widely used for modeling sentences. For instance, Li, Bing, Lam, and Shi (2018) employed LSTM to encode aspect words and context words respectively and then used the relatedness measure to link each context word with the associated aspect to create a completely interactive feature representation. Furthermore, RNN-based models are subsequently modified to incorporate the attention mechanism, which shows promise for performance improvement. Apart from the RNN network (Wang, Huang, Zhu, & Zhao, 2016), other network structures, such as CNN-based networks (Li et al., 2018; Xue & Li, 2018), memory networks (Chen, Sun, Bing and Yang, 2017; Tang, Qin, & Liu, 2016), and gated networks (Xue & Li, 2018) have also been investigated to integrate the attention mechanism. A representative work is Interactive Attention Network (IAN) (Ma, Li, Zhang, & Wang, 2017) learning the representation of aspect and context respectively by interactively calculating attention in the aspect and context. The attention mechanism allows the model to efficiently concentrate on the crucial portion of the text while ignoring the unnecessary ones, these important contents are viewed as aspect-related opinion words and expected to be assigned high weights. However, the weight of each word in the text is calculated based on the similarities of words. For the structured complex long text, non-aspect-related opinion words will inevitably be given high weights. As the example shown in Fig. 1, the first "poor" is the opinion word of the aspect "quality" and the latter "poor" is related to the aspect "performance". Existing attention-based methods will simply give both "poor" high weights since they are unable to distinguish which "poor" is the opinion word connected to the considered aspect.

Graph-based ALSC models: Researchers find that the dependency of a sentence can be viewed as graph-structured data (Phan et al., 2023; Sun, Zhang, Mensah, Mao and Liu, 2019; Zhang, Li, & Song, 2019), which reflects the geometric and topological relationships between words and solves the problem of the attention mechanism's inability to accurately determine opinion words. As a result, some works deliberately utilize the syntactical information and word dependencies of the sentence to seek more expressive sentence representation (Chen, Teng et al., 2022; Chen, Teng and Zhang, 2020; Gu, Zhao, He, Li, & Ying, 2023; Hou et al., 2021; Huang & Carley, 2019; Li, Chen et al., 2021; Tang, Ji, Li, & Zhou, 2020; Tian, Chen, & Song, 2021; Wang et al., 2020; Zhang & Qian, 2020; Zhou, Liao, Gao, Jie, & Lu, 2021; Zhu, Zhang et al., 2023). These works use the graph neural network (GNN) (Zhou, Cui et al., 2020) to model the dependency tree formed by dependency parsers. For example, Li, Chen et al. (2021) designed a SemGCN module with self-attention to capture semantic information and a SynGCN module to capture the syntactic knowledge to learn more accurate

dependency correlations between words. Similarly, Zhang et al. (2019) employed multiple GNN layers with an LSTM encoder to tackle the problems that previous models ignored. Although Graph-based methods can clarify the aspect-related opinion words and provide reasonable explanations for the prediction, they rely heavily on the correctness of the dependency structure of the sentence, which is another great challenge in the NLP community.

PLM-based ALSC models: Recently, NLP has entered the pre-trained era, where PLMs learn universal language representations on the large unannotated corpora and promote downstream tasks by a huge margin. Not surprisingly, PLMs have become a common building piece for the ALSC task (Dai et al., 2021; Hsu, Chen, Huang, & Chen, 2021; Sun, Huang and Qiu, 2019; Yong et al., 2023; Zhang, Zhang et al., 2022). For example, Li et al. (2019) demonstrated that utilizing a straightforward linear classification layer built on top of BERT can produce a performance that is more competitive than the earlier particularly created End-to-End ALSC models. PLMs can be regarded as a knowledge base (Petrone et al., 2019; Roberts, Raffel, & Shazeer, 2020), which stores knowledge in the model parameters, and downstream tasks can benefit by extracting relevant knowledge from PLMs implicitly. A large number of studies have shown that models with PLMs as the basic component perform best on ALSC by fine-tuning with domain data (Qiu et al., 2020). The powerful semantic expression of PLMs comes from their pre-training modes (Qiu et al., 2020), e.g. Language Modeling (LM) (Baevski, Edunov, Liu, Zettlemoyer, & Auli, 2019), Masked Language Modeling (MLM) (Devlin et al., 2019), Permuted Language Modeling (PLM) (Yang et al., 2019) and Denoising Autoencoder (DAE) (Lewis et al., 2020). Selecting or designing appropriate pre-training tasks for specific missions remains a challenge.

Knowledge-based ALSC models: To provide interpretability and boost accuracy, incorporating external knowledge into ALSC models is an effective approach (Jin et al., 2023; Wu et al., 2023; Zhong et al., 2023; Zhou, Huang et al., 2020). Moreover, with the establishment and application of enormous general-purpose knowledge bases, this approach becomes possible. As an illustration, Zhou, Huang et al. (2020) directly used the concept-level representation and semantic linkages offered by SenticNet (Cambria, Speer, Havasi, & Hussain, 2010) to enhance the connotations of aspect words and context words and, as a result, increase the accuracy of sentiment classification. Another recent example is the use of sets of WordNet (Miller, 1995) synonyms by Zhong et al. (2023) to represent ideas connected to sentence terms. By learning basic concepts first and progressing to more complex ones afterward, this method makes it easier to comprehend the meaning of sentences. However, using external information is not a free meal, it requires a significant amount of memory to store the necessary data, and accessing related knowledge has high computational costs. Additionally, adding external knowledge will introduce a lot of noise, which may hinder rather than help the performance improvement of the model.

In contrast to retrieving knowledge from external sources, we attempt to retrieve useful knowledge from the training data, which can be viewed as a further knowledge extraction and usage. The biggest drawback of retrieving knowledge from training data is that it also requires significant additional computation overhead. To alleviate this issue, the Memory bank technique is used to store the latest instance representations.

2.2. Retrieval technology

Recent research has shown that neural networks implicitly store the knowledge into model parameters during training and retrieve it during inference (Roberts et al., 2020). In other words, a neural model's performance is greatly impacted by the amount of visible data. As a result of the prohibitively expensive annotation expenses, adequate labeled data is, however, not available. This problem can be alleviated by using PLMs, which have gained universal knowledge from training on vast unlabeled corpora. Nevertheless, even a PLM as big as GPT-3 (Brown et al., 2020) cannot remember everything. It is important and necessary to provide a wealth of additional knowledge to the model. To achieve this, Information Retrieval (IR) technology is adopted to obtain relevant knowledge from outside sources and complete NLP tasks.

The two widely used IR techniques are sparse retriever and dense retriever where the frequently employed sparse retrievers are TF-IDF and BM25 (Robertson, Zaragoza, et al., 2009), they can effectively match keywords with an inverted index and are simple to modify for a generic large-scale search (Zhu, Xu et al., 2023). Recently, REINA (Wang et al., 2022) uses BM25 to retrieve the most relevant information from the training data to supplement the input. In addition, BM25 is also applied for open-domain question answering (Chen, Fisch, Weston and Bordes, 2017; Lin, Ji, Liu, & Sun, 2018; Wang et al., 2018) and machine translation (Gu, Wang, Cho, & Li, 2018). For the dense retriever, DPR (Karpukhin et al., 2020) has received the most research attention recently, whose retrieval scores are based on inner products between dense vectors, and it can retrieve semantically relevant examples in comparison to the sparse retriever.

As illustrated in Fig. 1, ideal retrieval instances are those that share the same aspects as the target instance, rather than those that have similar semantics. As a result, we first get retrieval instances by using the sparse retriever, which is more flexible than the dense retriever and has superior generalization and searching velocity. After that, for the aspect without enough retrieval instances, its semantically similar aspects can be found by the dense retriever, and then the lacking components will be supplemented by the retrieval instances of similar aspects. The process of obtaining the retrieval instances will be described in detail in Section 3.3 and exemplified in Section 5.3.1.

2.3. Contrastive learning

Contrastive Learning (CL) seeks to retrieve or generate comparable positive and negative instances for a target instance and learns the specific representation by reducing the distances between the target instance and positive instances while keeping negative instances distant from the target instance. Due to its superiority, CL has recently sparked a great deal of interest in the AI family,

particularly in the community of computer vision (CV) (Chen, Kornblith, Norouzi and Hinton, 2020; He, Fan, Wu, Xie, & Girshick, 2020), and it significantly improves the performance of neural models. From Inst Disc (Wu, Xiong, Yu, & Lin, 2018) to DINO (Caron et al., 2021), researchers integrated CL into the CV tasks thoroughly, and numerous cutting-edge technologies have been invented, which motivates the application of CL on the NLP tasks, including Recommendation (Elahi et al., 2023), Question Answering (Wilf, Ma, Liang, Zadeh, & Morency, 2023; Yang, Wei, Jiao, Jiang, & Yang, 2021), Text Classification (Xu, Zhang, Wu, & Wei, 2022).

The research lines of CL on NLP can be divided into two types (Hong, Zhang, Wang, & Zhao, 2022): (a) Improving the strategy for retrieving instances. High-quality positive and negative instances are of vital importance (Wu, Manmatha, Smola, & Krahenbuhl, 2017), and numerous studies have been done to improve the sample techniques used by CL (Xiong et al., 2021). A representative work is SimCSE (Gao, Yao, & Chen, 2021) which takes the entailment pairs of the NLI dataset as positive instances, and contradiction pairs as well as other in-batch instances as negative instances. (b) Developing better frameworks for CL. DensePhrase (Lee, Sung, Kang, & Chen, 2021), like MOCO (He et al., 2020), accesses more negative instances by using a memory bank without increasing GPU utilization. In addition, unlike DPR, the model achieves impressive performance by retrieving phrases rather than passages.

In our work, on the one hand, we achieve high-quality contrastive instances by retrieving instances with similar aspects from training data, where positive retrieval instances possess the same sentiment as the target instance at the same time. On the other hand, we ensure that there is at least one positive instance for the target instance by using retrieval instances, meanwhile, the number of negative instances for CL is increased. Furthermore, to effectively mine the knowledge of retrieval instances, we rank the retrieval instances according to their semantics with the target instance, where semantically dissimilar positive and semantically similar negative retrieval instances are preferentially used.

2.4. Similar works

Like Su et al. (2021), we do not develop a specific ALSC model but rather propose a training framework that is capable of enhancing the performance of existing ALSC models. Our proposed RCL framework is suitable for arbitrary existing ALSC model and does not need external computation, while the framework in Su et al. (2021) is restricted to attention-based models and requires additional training processes to find the attention weight of each instance. Another related work is GEN-SCL-NAT (Peper & Wang, 2022), which aims to solve the task of extracting Aspect-Category-Opinion-Sentiment (ACOS) quadruples. GEN-SCL-NAT reconstructs the instance representations by using Supervised Contrastive Learning (SCL), aiming to discriminable represent three sentiment characteristics: sentiment polarity, aspect term type, and opinion term type, where aspect-related characteristics are categorized into three taxonomies: has explicit aspects, has implicit aspects and has both. On the contrary, the aspect of each instance is provided explicitly in the ALSC task, and each aspect is regarded as a category in our work. Therefore, we categorize each instance according to the aspect itself rather than the aspect's characteristics, which is a more intuitive and granular way of learning the common features of similar instances.

3. Methodology

This section first reviews the definition and notations of ALSC, and then discusses each module of our suggested RCL framework in detail: Section 3.3 introduces how to get the retrieval instances; Section 3.4 introduces how to extract the common features of retrieval instances and make the ALSC model learn from the extracted features.

3.1. Task definition

A sentence $s = \{w_1, \dots, w_n\}$ toward a certain aspect $a = \{w_k, \dots, w_{k+n_a}\}$ is referred to as an instance, where n and n_a denote the word number of the sentence and aspect respectively. a is a span of s that satisfies $k \geq 1$ and $k + n_a \leq n$. ALSC aims to ascertain the sentiment polarities (e.g. "Negative", "Neutral", and "Positive") of instances (i.e. sentence s toward a), and the key point of most ALSC models lies in computing comprehensive representations of instances by different approaches.

3.2. Architecture overview

The overview of our proposed RCL is figured in Fig. 2, where an existing ALSC model is utilized as the encoder to generate the representations of instances. During the training processing, no parameters other than those of the ALSC model need to be updated. Hence, the proposed RCL framework works in a hot-plugging fashion, which is adaptable, flexible and compatible with almost all ALSC models. We will index the training instances when importing the training data. For the target instance, its retrieval instances are obtained through Elasticsearch, whose aspects are the same as that of the target instance. After that, we calculate the CL loss between the target instance and its retrieval instances to enforce positive retrieval instances gathered together in the feature space. Note that the representation of the target instance is generated in real-time while the representations of the retrieval instances are extracted from the Memory bank to avoid expensive calculation costs.

3.3. Retrieving instances from training data

Commonly, there are various descriptions of a product or service evaluation toward a certain aspect, which can also be observed in the public domain datasets, such as Restaurants and Twitter. Reviewing the example in Fig. 1, six "battery life" descriptions are displayed, which we called retrieval instances. We believe these descriptions (retrieval instances) have some similarities, which is good for ALSC. This section describes how to obtain the retrieval instances of a certain aspect.

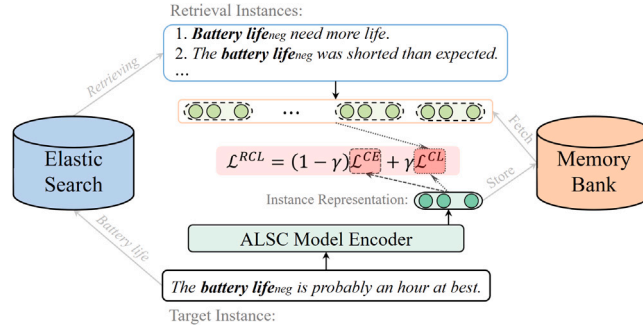


Fig. 2. Overview of retrieve contrastive learning framework, where implied two modules: (a) retrieving instances from training data; (b) learning common features from retrieval instances.

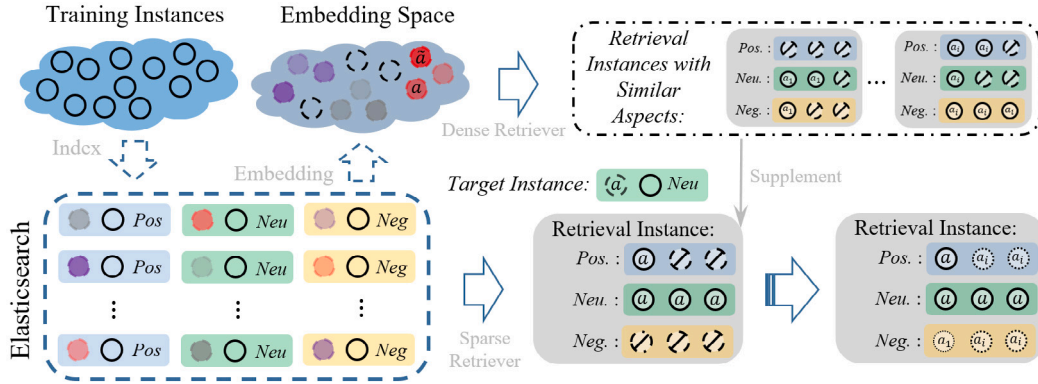


Fig. 3. Retrieving instances from training data.

3.3.1. Index the training instances

As illustrated in Fig. 3, we pre-process the training data and each instance constitutes a {key, value} pair, i.e. $d = (a, s)$ where s denotes the instance and a is the concerned aspect a . Next, to facilitate fast retrieval, we index training instances with the well-known full-text retrieval engine Elasticsearch (Amato, Bolettieri, Carrara, Falchi, & Gennaro, 2018), denoted as $\mathcal{E}$.

3.3.2. Retrieve instances by the sparse retriever

For a certain aspect a , its retrieval instances are defined as training instances that possess the aspect of a , which can be obtained by matching a with the aspect of other training instances (known as sparse retriever).

$$Ret_a = \mathcal{E}(a \mid A) \quad (1)$$

where A is the set of all training instances' aspects, and Ret_a denotes the set of retrieval instances of aspect a that include all the training instances with a as their aspects. For the target instance $d = (a, s)$, its retrieval instances set is a subset of Ret_a , defined as:

$$Ret_d = Ret_a - d \quad (2)$$

where Ret_d denotes the retrieval instance set of instance d , the difference between Ret_d and Ret_a is whether d is included or not.

Furthermore, we limit the number of retrieval instances to k per sentiment category. Hence, as illustrated in Fig. 4, the set of retrieval instances Ret_d for a given instance $d = (a, s)$ is one of the following four cases ($k = 3$ for example):

- (a) Ret^{no} : the retrieval instances are null.
- (b) Ret^{few} : the retrieval instances are not null but at least one category for which the retrieved instance is empty.
- (c) Ret^{some} : the retrieval instances of all categories are not null but at least one category for which the retrieved instance is not full.
- (d) Ret^{ample} : the retrieval instances of all categories are full.

For the first three cases, we need to supplement the retrieval instances with the dense retriever.

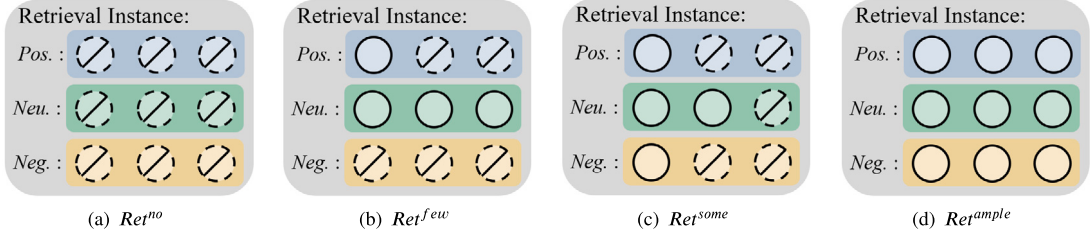


Fig. 4. Retrieval instances obtained by the sparse retriever.

3.3.3. Supplement retrieval instances by the dense retriever

Inspired by Sato, Yamada, and Kashima (2022), we use SBERT (Reimers & Gurevych, 2019) to discover semantically similar aspects. As depicted in Fig. 3, all aspects are embedded into the vector space, where semantically similar aspects are naturally closer to one another, such as “staff” and “wait staff”. To supplement the incomplete retrieval instances set Ret_d , we first look for the most similar aspects A_a and then use their retrieval instances Ret_{A_a} to supplement the target aspect’s retrieval instances until they are sufficient for all sentiment categories.

$$\tilde{A}_{-a} = \text{SBERT}_{\text{sim}}(a, A_{-a}) \quad (3)$$

where A_{-a} denotes all pre-defined aspects except for a , $\text{SBERT}_{\text{sim}}$ is the SBERT-based similarity measure function that returns the ranked aspects $\tilde{A}_{-a}$ that is most similar to a . As shown in Fig. 3, we iterate over the most similar aspect $\tilde{a} \in \tilde{A}_{-a}$ and supplement Ret_d with its retrieval instances $Ret_{\tilde{a}}$ until all categories of Ret_d are full.

$$Ret_d = Ret_d + Ret_{\tilde{a}} \quad (4)$$

where $Ret_{\tilde{a}}$ denotes the retrieval instances set of a similar aspect $\tilde{a}$.

3.3.4. Rank the retrieval instances

Based on the presumption that semantically dissimilar positive retrieval instances and semantically similar negative retrieval instances relative to the target instance are more valuable, we rank the retrieval instances to prioritize the utilization of valuable retrieval instances. We first distinguish the positive and negative retrieval instances according to their sentiment polarity.

$$\begin{cases} Ret_d^{\text{pos}} \Leftarrow \tilde{d} \in Ret_d \text{ if } y_{\tilde{d}} = y_d \\ Ret_d^{\text{neg}} \Leftarrow \tilde{d} \in Ret_d \text{ if } y_{\tilde{d}} \neq y_d \end{cases} \quad (5)$$

where y_d and $y_{\tilde{d}}$ denotes the sentiment polarity of instance d and $\tilde{d}$ respectively. Ret_d^{pos} and Ret_d^{neg} denote the sets of positive and negative retrieval instances respectively. As in the case of Fig. 1, for the target instance with the aspect of “battery life” and sentiment of “Positive”, there are six retrieval instances with the first three being positive retrieval instances and the last three being negative retrieval instances (one with “Neutral” sentiment and two with “Negative” sentiments).

And then, we use the same similarity measure function $\text{SBERT}_{\text{sim}}$ to calculate the similarities of the target instance and its retrieval instances. Therefore, semantically similar and dissimilar retrieval instances are ascertained accordingly.

$$\begin{cases} \widetilde{Ret}_d^{\text{pos}} = \text{SBERT}_{\text{sim}}(d, Ret_d^{\text{pos}}) \\ \widetilde{Ret}_d^{\text{neg}} = \text{SBERT}_{\text{sim}}(d, Ret_d^{\text{neg}}) \end{cases} \quad (6)$$

where $\text{SBERT}_{\text{sim}}$ denotes sort in ascending, that is, return the least similar instances with the target instance. $\widetilde{Ret}_d^{\text{pos}}$ and $\widetilde{Ret}_d^{\text{neg}}$ are ranked positive and negative retrieval instances, respectively.

3.4. Learn from retrieval instances

We will describe how to use retrieval instances to enhance the performance of ALSC models in this section.

3.4.1. Instance encoding

As shown in Fig. 2, RCL needs to integrate an ALSC model as the Encoder to generate the representations of instances. For each instance d , its representation vector h is obtained:

$$h = \text{Encoder}(d) \quad (7)$$

Meanwhile, representations of retrieval instances can also generate by the Encoder, however, doing so will incur significant computational costs. Instead, the Memory bank (Wu et al., 2018) ($\mathcal{M}$) is utilized to avoid additional intensive computing. During

the training processing, we store this representation vector h in the Memory bank $\mathcal{M}$ ($h \Rightarrow \mathcal{M}$). At the same time, we can retrieve the retrieval instances' representation vectors from the Memory bank $\mathcal{M}$ based on the ranked retrieval instances.

$$H_{Ret} = \begin{cases} H_{Ret}^{pos} \Leftarrow \mathcal{M}[\widetilde{Ret}_d^{pos}] \\ H_{Ret}^{neg} \Leftarrow \mathcal{M}[\widetilde{Ret}_d^{neg}] \end{cases} \quad (8)$$

where $\mathcal{M}[*]$ denotes taking out the specified instance representation from the memory bank. H_{Ret} denotes the set of retrieval instances representations correspondingly, which consists of the set of positive retrieval instances representations H_{Ret}^{pos} and the set of negative retrieval instances representations H_{Ret}^{neg} .

3.4.2. Retrieval contrastive learning

CL seeks to learn effective representation by grouping the positive instances and severing apart the negative instances (Hong et al., 2022; Khosla et al., 2020), which is an effective method of learning across instances. To learn from retrieval instances, CL is undoubtedly a direct and effective way. Hence, we follow the loss calculation in Gao et al. (2021), which determines the contrastive loss of each instance in a batch and is expressed as:

$$\mathcal{L}^{CL} = -\log \frac{\sum_{h^+ \in H^+} e^{\text{sim}(h, h^+)/\tau}}{\sum_{h' \in (H^+ \cup H^-)} e^{\text{sim}(h, h')/\tau}} \quad (9)$$

where $\text{sim}(*, *)$ denotes the cosine similarity function. τ is a temperature hyperparameter. H^+ and H^- are the positive and negative instances set respectively.

The positive and negative instances used for CL are different for different tasks and datasets. In our setting, we use the positive retrieval instances as H^+ , the negative retrieval instances and in-batch negative sampling as H^- . Specifically, for each target instance h , there are $3k$ retrieval instances in H_{Ret} , including k instances in H_{Ret}^{pos} and $2k$ instances in H_{Ret}^{neg} , where k is a hyperparameter of RCL. Hence, for an instance h in a mini-batch H_{batch} , its positive and negative instances' representations are:

$$H^+ = H_{Ret}^{pos} \quad (10)$$

$$H^- = H_{Ret}^{neg} + H_{batch} - h \quad (11)$$

In this way, the common features of retrieval instances can be extracted subtly. On the one hand, we provide higher quality positive samples for CL; on the other hand, we greatly increase the number of negative samples, which is proved to be good for CL.

3.4.3. Loss functions of RCL

Following the basic paradigm of cross-entropy loss, firstly, the probability distribution of the predicted sentiment polarity is generated through a linear classifier followed by a softmax function:

$$\hat{y} = \text{softmax}(Wh + b) \quad (12)$$

where W and b are the parameters of the classifier. And then, the classification loss is calculated:

$$\mathcal{L}^{CE} = -y \log(\hat{y}) \quad (13)$$

where y is the ground truth.

RCL aims to use CL to learn the common features in retrieval instances to enhance the representation ability of the existing ALSC model. Therefore, the general training objective of RCL is to minimize the combination of CL loss and classification loss:

$$\mathcal{L}^{RCL} = (1 - \gamma)\mathcal{L}^{CE} + \gamma\mathcal{L}^{CL} \quad (14)$$

where $0 < \gamma < 0.5$ is another hyperparameter of RCL that estimates the importance of the CL loss.

During the inference stage, we only use Eq. (12) to obtain the probability distribution of each inference instance, and the classification result with the highest probability is taken as the result of sentiment polarity prediction.

4. Experimental setup

4.1. Datasets

We run tests on three widely used datasets that are released in public, where Restaurants and Laptops were made by Pontiki, Galanis, Papageorgiou, Manandhar, and Androutsopoulos (2015), and Twitter was a collection of tweets (Dong et al., 2014). All datasets are accessible through the Pytorch-ABSA project,¹ where each sentence is annotated with aspects and their associated sentiment polarities ("Positive", "Negative", and "Neutral"). The statistics of three datasets are summarized in Table 1, where the last column depicted the ratio of aspects with more than one vanilla retrieval instance. As shown in Table 1, most aspects on Twitter

¹ <https://github.com/songyouwei/ABSA-PyTorch/tree/master/datasets>.

Table 1
Statistics of the three benchmark datasets.

Dataset	Type	Positive	Negative	Neutral	Rate
Laptops	Train	994	870	464	26.77%
	Test	341	128	169	
Restaurants	Train	2164	807	637	24.79%
	Test	727	196	196	
Twitter	Train	1561	1560	3127	79.65%
	Test	173	173	346	

possess multiple training instances, which means that when retrieving instances for the target instances, the retrieval instances are more likely to have the same aspects as the target instance. Compared with retrieval instances that have similar aspects as the target instance, retrieval instances with the same aspect are considered high-quality retrieval instances. In contrast, only about a quarter of the aspects on Laptops and Restaurants have more than one training instance, which inevitably results in the retrieval instances of the target instance mainly derived from the dense retriever. In other words, most of the retrieval instances in Laptops and Restaurants are those with similar aspects rather than with the same aspect as the target instance. Hence, the retrieval instances provided by Twitter are of higher quality than those provided by Laptops and Restaurants.

4.2. Baseline models

To illustrate the effectiveness of our proposed RCL framework, five ALSC models are used for comparison.

- **MemNet** (Tang et al., 2016), the lightweight but well-performing model, is one of the most classical ALSC models in the ABSA community, which explicitly reports the importance of each context word by multiple attention modules over the outside memory component.
- **TNet** (Li et al., 2018) obtains contextualized representations of context by LSTM, and then, the information of aspect and context are fused fully by the CPT module, which is an attention-based method. Finally, a CNN layer is followed to extract salient features for sentiment prediction.
- **ACLT²** (Zhou et al., 2021) is a graph-based ALSC method that can learn an aspect-centric latent tree with a root refinement strategy to better correlate the aspect and opinion words than the standard parse tree. Experiment results demonstrate that ACLT can better aggregate opinion information by shortening the distance between aspects and their corresponding opinion words.
- **Paraphrase³** (Zhang et al., 2021) cast the aspect sentiment quad prediction (ASQP) task to a paraphrase generation process, which inputs a review sentence and outputs sentiment quads that exists in the sentence. In our setting, we redesigned the instance output target based on the known aspect information to fit the ALSC task. For example, an input-target pair (x, y) is constructed as (The sentiment of “battery life” in “The battery life is probably an hour at best”. is, Negative).
- **ABSA-DeBERTa⁴** (Silva & Marcacini, 2021) is an advanced model in the ALSC community, which obtained the state-of-the-art results using the DeBERTa (Decoding-enhanced BERT with Disentangled Attention) as the feature encoder.

We used Macro-F1 and Accuracy as our evaluation metrics. Notably, the official codes of MemNet and TNet are not Pytorch-based. As an alternative, public models from the project Pytorch-ABSA⁵ are employed.

4.3. Model parameters setting

In the implementation, we first execute parameter searching for ALSC models to achieve their best performance as baselines. And then, we retrain these models with the same training parameters and random seeds in the RCL framework, which ensures that outcomes are the same as baselines when the retrieval instances are not used and allow us to intuitively investigate the impact of retrieval instances on ALSC. The hyperparameters for each model are tabulated in Table 2, where γ in Eq. (14) is one of the two parameters of RCL. The other parameter of RCL, the number of retrieval instances k mentioned in Sections 3.3.2 and 3.4.2 is empirically set as 3 on Laptops, 3 on Restaurants and 1 on Twitter, respectively.

² <https://github.com/zyxnlp/aclt>.

³ <https://github.com/isakzhang/absa-quad>.

⁴ <https://github.com/huberemanuel/DeBERTa/>.

⁵ <https://github.com/songyouwei/ABSA-PyTorch/tree/master/models>.

Table 2
Hyperparameters for retraining ALSC models with RCL.

	MemNet	TNet	Paraphrase	ACLT	ABSA-DeBERTa
Batch size	64	64	16	16	16
Early stop	10	10	3	4	4
max epochs	64	64	10	16	16
Learning rate	1e-1	1e-1	3e-4	3e-5	3e-5
Warmup rate	–	–	0.3	0.3	0.3
Optimizer	SGD	SGD	AdamW	AdamW	AdamW
Pre-train	Glove	Glove	T5	BERT	DeBERTa
Dropout	0.3	0.3	0.3	0.3	0.3
γ (RCL)	0.1	0.1	0.1	0.1	0.3

Table 3
Enhanced ALSC models by the RCL framework.

Models	Laptops		Restaurants		Twitter	
	F1	Acc	F1	Acc	F1	Acc
MemNet	67.50	72.41	70.13	80.00	70.33	71.39
MemNet+RCL	68.37 _{10.87}	73.04 _{10.63}	70.85 _{10.72}	79.20 _{10.80}	71.41 _{11.08}	72.11 _{10.72}
TNet	71.56	75.71	69.90	79.29	68.59	70.66
TNet+RCL	72.77 _{11.21}	76.33 _{10.62}	71.25 _{11.35}	78.93 _{10.36}	69.10 _{10.51}	70.95 _{10.29}
ACLT	75.80	79.47	78.61	85.18	73.53	74.71
ACLT+RCL	76.35 _{10.55}	79.94 _{10.47}	79.83 _{11.22}	86.34 _{11.16}	75.09 _{11.56}	75.72 _{11.01}
Paraphrase	77.98	81.82	80.67	87.32	73.95	75.29
Paraphrase+RCL	79.51 _{11.53}	82.29 _{10.47}	82.77 _{12.10}	88.48 _{11.16}	74.47 _{10.52}	75.00 _{10.29}
ABSA-DeBERTa	79.41	82.29	82.34	88.39	74.11	74.71
ABSA-DeBERTa+RCL	80.28 _{10.87}	82.76 _{10.47}	84.68 _{12.34}	89.38 _{10.99}	77.47 _{13.36}	78.32 _{13.61}

5. Experimental results

5.1. Enhancing ALSC models with RCL

5.1.1. Main results

The experimental results for evaluating the capability of RCL are tabulated in Table 3. Note that, for a fair comparison, the result of each retrained model with RCL is the same as the baseline when $\gamma = 0$. From Table 3, it can be found that the performance of each model is enhanced by training with RCL, which indicates that RCL enables the model to learn useful aspect-related knowledge from the retrieval instances and then polish the representation of the target instance for better sentiment prediction.

Specifically, MemNet is good at modeling key information in sentences and avoiding the interference of redundant information and improved significantly by introducing additional useful knowledge, where Macro-F1 score increased by 0.87% on Laptops, 0.72% on Restaurants and 1.08% on Twitter, respectively. The performance improvement on Twitter is more pronounced probably due to Twitter's ability to provide higher-quality retrieval instances (as shown in Section 4.1), and thus better meet the needs of MemNet for external knowledge. TNet also achieves great performance improvement (1.21% on Laptops, 1.35% on Restaurants and 0.51% on Twitter) by training with RCL. From the model framework (Li et al., 2018), we can find that TNet incorporates the aspect information into context words, which brings a deep understanding of aspects. Connecting the instance aspect and its similar aspects can further enhance the understanding of aspects. Hence, the performance improvements of TNet on Laptops and Restaurants are more significant since more retrieval instances with similar aspects are provided.

For the PLM-based models (ACLT and ABSA-DeBERTa) a wealth of language knowledge is stored in the model parameters, and fine-tuning mainly activates pathways that extract useful knowledge for domain tasks. Therefore, the larger the training set size, the better the PLM model can fit the domain data distribution. The retrieval instances in RCL are the reuses of the training instance in different ways, which enhances the data scale to some extent. After training with RCL, ACLT achieves 0.55%, 1.22% and 1.56% improvements in Macro-F1 on Laptops, Restaurants and Twitter, respectively, whereas ABSA-DeBERTa's performance improvements (0.87% on Laptops, 2.34% on Restaurants and 3.36% on Twitter) are about double that of ACLT. Note that, ACLT is also a graph-based method that models the important part of the sentence with the help of the dependency tree. Similar to MemNet, ACLT should theoretically have a further great performance gain trained with RCL on Twitter, but we do not observe this happening in our experimental results. A possible conjecture is that the ability of ACLT mainly comes from PLM.

Paraphrase is a generated framework that detects all sentiment elements jointly for a given opinionated sentence, i.e. aspect term, aspect category, opinion term, and sentiment polarity, while ALSC is a fine-grain task that aims to identify the sentiment polarity for a given sentence toward the concerned aspect. For a fair comparison, we redesign the input-output paradigm of Paraphrase to adapt the ALSC task. Compared with classification methods, the generation space of the generated-based method is very large, and the small-scale training set cannot well reduce the scope of generation space, so that leaves a lot of room for improvement. Therefore, Paraphrase trained with RCL has a large performance improvement (Macro-F1 score: 77.98% $\rightarrow$ 79.51%) on Laptops with a small amount of data, while the improvement (Macro-F1 score: 73.95% $\rightarrow$ 74.47%) is not obvious to Twitter with a large amount of data.

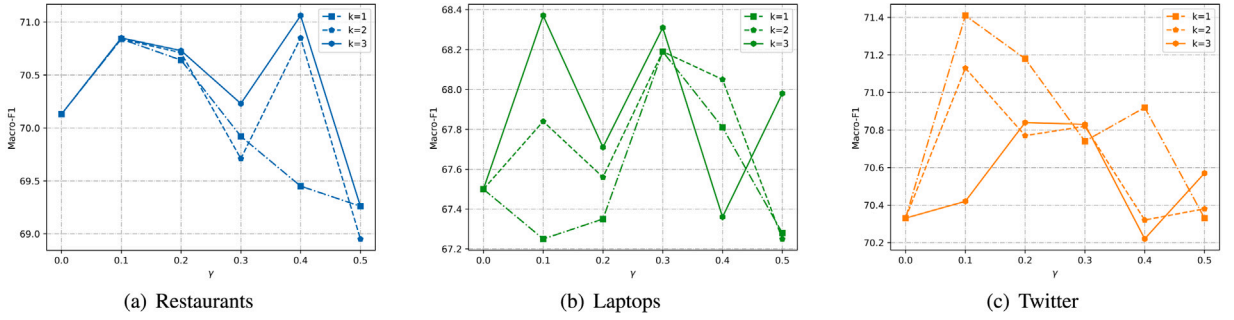
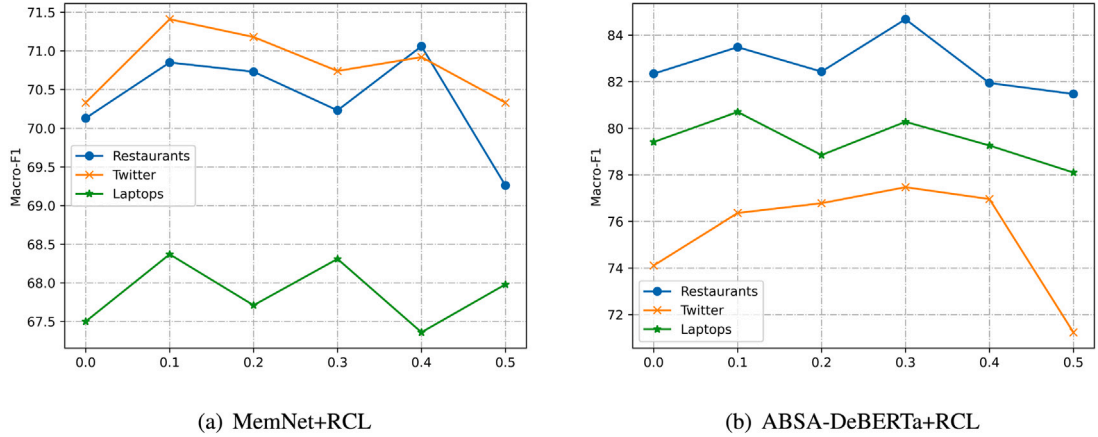


Fig. 5. RCL parameters searching on MemNet.

Fig. 6. Effect of the number of retrieval instance k .

5.1.2. Effect of RCL hyperparameters

Two hyperparameters need to be determined when applying our proposed RCL:

- The number of retrieval instances k , a dataset-related hyperparameter, confirms the most efficient knowledge contained by the minimum number of retrieval instances. To this end, we perform a grid search on MemNet, with k varying from 1 to 3 and γ varying from 0.0 to 0.5. Experimental results shown in Fig. 5 suggest that setting k to 3 on Laptops, 3 on Restaurants and 1 on Twitter are good choices, which corresponds to the statistic of aspects in Table 1.
- The CL loss coefficient γ , a model-related hyperparameter, reflects how the model best fits into RCL. Based on the selection of k , we conduct some exploratory experiments on MemNet and ABSA-DeBERTa to investigate the effects of γ in CL, with γ varying from 0.0 to 0.5. Experimental results are illustrated in Fig. 6, where Fig. 6(a) indicates that MemNet obtain the best results when γ setting to 0.1 on Laptops, 0.4 on Restaurants and 0.1 on Twitter, respectively. Fig. 6(b) demonstrates that ABSA-DeBERTa achieves the maximum performance enhancement when γ sets to 0.3 on all datasets. Therefore, as shown in Table 2, we set γ equal to 0.1 and 0.3 for MemNet and ABSA-DeBERTa, respectively. Different models possess their own optimal γ shown in Table 2 that are determined in the same way above.

5.1.3. Effect of different dense retrievers

We conduct some exploratory experiments to investigate the impact of the use of different dense retrievers when obtaining retrieval instances. The experimental results on ABSA-DeBERTa are figured in Fig. 7, where the SBERT-based retriever outperforms the Glove-based retriever in all datasets when γ is less than 0.4, while the Glove-based retriever is better when γ equals 0.5. We will perform a case study in Section 5.3.1 to show the differences between retrieval instances obtained by different dense retrievers.

5.1.4. Ablation studies

We conduct ablation studies to investigate the impact of the use of negative retrieval instances or not. The experimental results on ABSA-DeBERTa illustrated in Fig. 8 indicate that model performance is more stable when negative retrieval instances are not used. Even though the use of negative retrieval instances results in performance changes drastically, they can benefit the model to achieve the best performance. We guess that there exists a lot of helpful knowledge and noise information in the negative retrieval

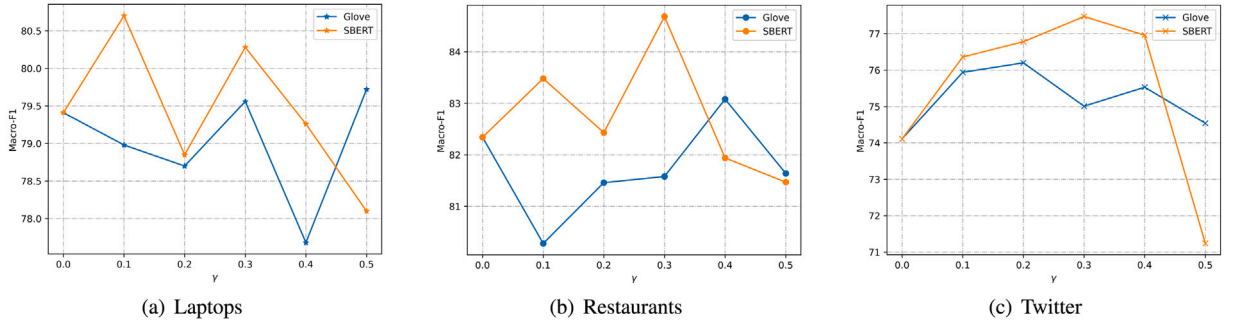


Fig. 7. Impact of different dense retrievers.

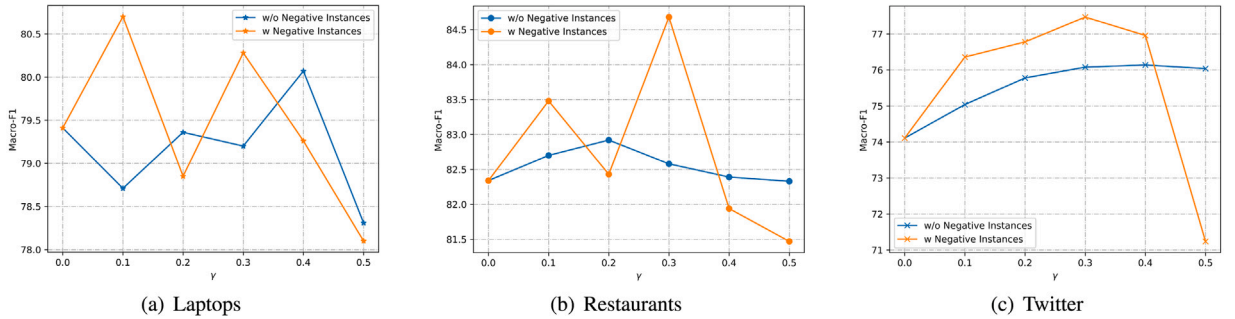


Fig. 8. Impact of the use of negative retrieval instances.

instances. Finding the best way to match the model and RCL to effectively extract the effective information from the negative retrieval instances can greatly improve the performance of the model.

In addition, as shown in Table 4, the performance of DeBERTa-RCL drops dramatically on all datasets when the negative retrieval instances are not used during training, which indicates that DeBERTa-RCL can filter noise and accurately extract useful knowledge from negative retrieval instances to enhance the target instance representation for sentiment prediction.

5.2. Performance comparison with advanced methods

Table 4 shows the comparisons between DeBERTa-RCL and the current cutting-edge methods, where SyMux (Fei et al., 2022), GEN (Yan, Dai, Ji, Qiu, & Zhang, 2021) and Dual-MRC (Mao, Shen, Yu, & Cai, 2021) are designed for joint sentiment analysis tasks, and we only extract the results of the ALSC task from the original paper. From Table 4, it can be found that the joint sentiment analysis task frameworks are competitive even though they are not designed separately for the ALSC task, where SyMux achieves the second-best result on the Restaurants and outperforms most of the BERT-base models on the Laptops. The DeBERTa-base ALSC model outperforms all BERT-base ALSC models, in particular, the Macro-F1 score increases by 0.87% on Restaurants, which demonstrates that DeBERTa is superior to BERT with respect to text semantic understanding. Hence, we use DeBERTa-base⁶ as the backbone and train the model with RCL, dubbed DeBERTa-RCL. Compared with ABSA-DeBERTa, DeBERTa-RCL can generate semantically rich instance representations and obtain 0.92% and 1.26% in terms of Macro-F1 improvement on Laptops and Restaurants, respectively. In addition, DeBERTa-RCL achieves new state-of-the-art results on Twitter, which indicates that linking instances with their aspect similar instances can enhance instance semantics to improve the accuracy of the ASLC task. We will perform the case study later and further analyze the principle of RCL.

5.3. Case study

5.3.1. Obtain retrieval instances

To better understand the behavior of retrieving instances from training data, we exemplify a case with the target instance “For great **Chinese food**_{pos} nearby, you have Wu Liang Ye and Grand Sichuan just a block away.”, and its retrieval instances ($k = 3$) are shown in Table 5, where “ $a : value$ ” denotes the similarity between the aspect of the retrieval instance and that of the target instance, and “ $s : value$ ” denotes the similarity between the retrieval instance and the target instance. The first three rows are

⁶ <https://github.com/huberemmanuel/DeBERTa/>.

Table 4

Experimental results comparison of advanced methods on the three datasets.

PLM	Methods	Laptops		Restaurants		Twitter	
		F1	Acc	F1	Acc	F1	Acc
RoBERTa-base (170M)	SyMux (Fei et al., 2022) ^a	78.99	–	84.45	–	–	–
BART-large (467M)	GEN (Yan et al., 2021) ^a	76.76	–	75.56	–	–	–
BERT-large (335M)	Dual-MRC (Mao et al., 2021) ^a	75.97	–	82.04	–	–	–
BERT-base (109M)	SPM (Hsu et al., 2021)	–	77.93	–	81.50	–	–
	ACLT (Zhou et al., 2021)	75.83	79.68	78.44	85.71	74.51	75.48
	DualGCN (Li, Chen et al., 2021)	78.10	81.80	81.16	87.17	76.02	77.40
	DR-BERT (Zhang, Zhang et al., 2022)	78.16	81.45	82.31	87.72	76.10	77.24
	dotGCN (Chen, Teng et al., 2022)	78.10	81.03	80.49	86.16	77.00	78.11
	EK-GCN (Gu et al., 2023)	79.19	81.30	82.55	87.65	75.16	75.89
DeBERTa-base (139M)	ABSA-DeBERTa (Silva & Marcacini, 2021)	79.36	82.76	83.42	89.46	–	–
	DeBERTa-RCL	80.28	82.76	84.68	89.38	77.47	78.32
	DeBERTa-RCL _{-Neg}	79.20 _{11.08}	81.82 _{10.94}	82.58 _{12.10}	88.39 _{10.99}	76.08 _{11.39}	77.60 _{10.72}

^a Denotes the method is not designed separately for ALSC, others are all ALSC models.

Table 5

An Example of retrieving instances for the target instance by using two dense retrievers.

Target instance: For great Chinese food_{pos} nearby, you have Wu Liang Ye and Grand Sichuan just a block away.			
SBERT-based dense retriever		Glove-based dense retriever	
Retrieval instances ($k = 3$)	Sim.	Retrieval instances ($k = 3$)	Sim.
We actually left hungry and went across the street to Wo Hop at 15 Mott Street for some good Chinese food_{pos} .	$a : 1.00$ $s : 0.56$	We actually left hungry and went across the street to Wo Hop at 15 Mott Street for some good Chinese food_{pos} .	$a : 1.00$ $s : 0.90$
Next time, we wouldn't dare order anything else other than some simple Asian appetizers_{pos} and drinks.	$a : 0.66$ $s : 0.56$	Decent Thai food_{pos} in cute-though a bit dank-little Nolita hangout, but service terrible.	$a : 0.88$ $s : 0.82$
This place, however, has a lot less pretension than Joya and the Thai food_{pos} is still above average.	$a : 0.62$ $s : 0.47$	Spice is great Thai food_{pos} , love the inexpensive appetizers.	$a : 0.88$ $s : 0.84$
If you want Americanized Chinese food_{neu} with your usual watery, generic white sauce, this is your place.	$a : 1.00$ $s : 0.63$	If your favorite Chinese food_{neu} is general tao chicken, then this is not your place.	$a : 1.00$ $s : 0.89$
If your favorite Chinese food_{neu} is general tao chicken, then this is not your place.	$a : 1.00$ $s : 0.57$	This isn't your corner Chinese food_{neu} takeout.	$a : 1.00$ $s : 0.88$
This isn't your corner Chinese food_{neu} takeout.	$a : 1.00$ $s : 0.54$	If you want Americanized Chinese food_{neu} with your usual watery, generic white sauce, this is your place.	$a : 1.00$ $s : 0.87$
They even have a section in the menu called American Chinese food_{neg} !	$a : 0.87$ $s : 0.60$	They even have a section in the menu called American Chinese food_{neg} !	$a : 0.95$ $s : 0.87$
Don't eat here unless you're starving for Thai food_{neg} and you work next door.	$a : 0.62$ $s : 0.48$	Don't eat here unless you're starving for Thai food_{neg} and you work next door.	$a : 0.88$ $s : 0.86$
Plain and simple it's bad Thai food_{neg} .	$a : 0.62$ $s : 0.43$	For many people, this may not seem like Authentic Thai food_{neg} because most places in NYC aren't quite authentic.	$a : 0.88$ $s : 0.86$

positive retrieval instances that have the same sentiment polarities as that of the target instance and the last six rows are negative retrieval instances. Retrieval instances with the same aspects as the target instance are retrieved by the sparse retriever, and others are retrieved by the dense retriever.

From Table 5, it can be found that the Glove-based retriever tends to retrieve homotopic words of the target aspect (all retrieval aspects have the word “food”), while the SBERT-based retriever can retrieve aspects with similar semantics but different characters, such as “Asian appetizers”. In addition, the value of similarity calculated by the SBERT-based retriever is significantly lower than that calculated by the Glove-based retriever when inputting the same instance pair, such as the first positive retrieval instance in Table 5 ($s_{SBERT} : 0.56 < s_{Glove} : 0.90$), which indicates SBERT has a higher requirement for similar text determination. Furthermore, take “This isn't your corner **Chinese food_{neu}** takeout.” and “If you want Americanized **Chinese food_{neu}** with your usual watery, generic white sauce, this is your place.” as an example, the former talks about takeaways and the latter about restaurants. Contrary to the SBERT-based retriever, the glove-based retriever considers the former to be closer to the target instance than the latter. Therefore, the SBERT-based retriever outperforms the Glove-based retriever from this point of view since the target instance is talking about the restaurant rather than takeaways.

Table 6
Case study on several test instances.

Type	Instances	Truth	Baseline	RCL
Ret^{no}	<i>The portions of the food_{neu} that came out were mediocre.</i>	Neutral	Negative ⊗	Neutral ✓
	<i>New hamburger with special sauce_{pos} is ok - at least better than big mac!</i>	Positive	Neutral ⊗	Positive ✓
	<i>Desserts include flan and sopaipillas_{neu}.</i>	Neutral	Positive ⊗	Neutral ✓
Ret^{few}	<i>For someone who used to hate Indian food_{pos}, baluchi's has changed my mind.</i>	Positive	Negative ⊗	Positive ✓
Ret^{some}	<i>Pizzas were excellent in addition to appetizers_{pos} and main courses.</i>	Positive	Neutral ⊗	Positive ✓
	<i>A beautiful atmosphere, perfect for drinks and/or appetizers_{neu}.</i>	Neutral	Positive ⊗	Neutral ✓
Ret^{ample}	<i>For the price you pay for the food_{neg} here, you'd expect it to be at least on par with other Japanese restaurants.</i>	Negative	Positive ⊗	Negative ✓
	<i>Although the restaurant itself is nice, I prefer not to go for the food_{neg}.</i>	Negative	Neutral ⊗	Negative ✓
	<i>This place has the strangest menu and the restaurants try too hard to make fancy food_{neg}.</i>	Negative	Neutral ⊗	Negative ✓

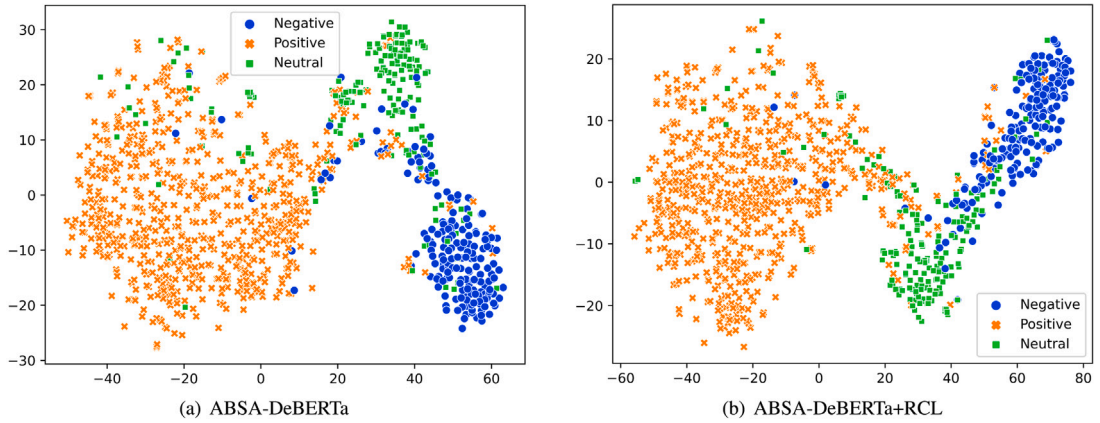


Fig. 9. The t-SNE visualization results of ABSA-DeBERTa with and without trained with RCL on Restaurants.

5.3.2. Performance improvement with RCL

To know how our proposed RCL improves neural ALSC models, we show the comparison between predictions of baseline (ABSA-DeBERTa) and model trained with RCL (ABSA-DeBERTa+RCL) on several test instances in Table 6. These instances are categorized into four categories according to the type of retrieval instances obtained by the sparse retriever, as shown in Section 3.3.2.

First, for instances with unseen aspects, the baseline model is unable to understand aspects that have not been trained. Take the structure simple sentence “Desserts include flan and *sopaipillas_{neu}*.” as an example, this sentence simply states a fact and belongs to “Neutral” sentiment obviously, while the baseline thinks it is “Positive”. Hence, one of the advantages of our method is that a model trained with RCL can understand any aspect term by linking it with semantically similar aspects that have been well-trained. Second, for instances with not well-trained aspects, the baseline model is easily suffered from the noise information. For example, in the sentence “For someone who used to hate *Indian food_{pos}*, *baluchi's* has changed my mind.”, ABSA-DeBERTa is confused by the adjacent sentiment word “hate” and predicts it is “Negative” toward the aspect of “*Indian food*”. Thus, another advantage of our method is that RCL enables the model to filter noise information and obtain correct sentiment prediction. Finally, for instances with well-trained aspects (such as “food”), the baseline model cannot extract the implicit opinion and result in incorrect sentiment prediction, especially for the long sentence. When training with RCL, instances with implicit opinions will link to similar instances with explicit opinions, and instances with explicit opinions will also link to similar instances with implicit opinions, which enables the model to be aware of implicit opinions in the sentence and generate richer semantic representations for sentiment prediction.

Furthermore, we visualized the distribution of test instances of Restaurants by using t-SNE (Van der Maaten & Hinton, 2008), shown in Fig. 9. From Fig. 9, we see the baseline model trained with RCL further gathers instances with different sentiments and makes the representation space more compact. An obvious phenomenon is that most instances with “Negative” sentiment polarities that are scattered in the “Positive” region have been gotten back to the “Negative” region.

6. Conclusion

Generating a refined instance representation is an essential requirement of the ALSC task, which demand eliminating redundant information while increasing the useful information as much as possible. To this end, we propose an RCL framework that enables

the existing ALSC models to generate polished instance representations while preserving their advantages at the same time. RCL consists of two modules and works in a pipeline fashion. In the first stage, we conduct data mining using the aspect information, aiming to discover instances that share the same aspects by using a sparse retriever and a dense retriever. In the second stage, we use CL to mine the intrinsic connections between the target instance and its retrieval instances. Retrieval instances are versatile, which bring additional useful information and help to shield the redundant information. Extensive experimental results reveal that the ALSC model can be enhanced by being trained with RCL, which demonstrates the superiority of the proposed RCL framework.

RCL is adaptable, flexible, and compatible with the current ALSC models, and the two main tenets of RCL are how to acquire more precise retrieval instances and how to efficiently extract the common features present in the retrieval instances. We will continue to study these issues in the future.

7. Discussion

7.1. The application of RCL

RCL is a training framework that utilizes an existing ALSC model as the backbone without introducing additional training parameters. In other words, RCL post-processes the instance representations generated by the original ALSC models, hence, the advantages of the original ALSC models are retained. For each target instance, RCL narrows the distances between it and its positive retrieval instances in the embedding space by using CL while increasing the distances of negative instances to polish the instance representation. Therefore, the essence of RCL is to further refine the representation of the target instance by the guidance of its retrieval instances. In the inference stage, the original paradigm of the ALSC model is followed that the instance representations are generated by the trained model, and then the predicted results are output after passing through the classifier.

7.2. The limitations of RCL

RCL is a method of refining instance representations, whose performance heavily depends on the quality of the retrieval instances. In the ALSC task, high-quality retrieval instances are obtained by retrieving from the training data according to instances' aspect information. Hence, RCL is not suitable for joint ABSA tasks that did not provide aspect terms explicitly, such as Aspect term Extraction (AE), Triplet Extraction (TE) (Fei et al., 2022). In addition, retrieval instances are only used when training the ALSC model to enable the model to generate polished instance representations. In the inference stage, we cannot fully distinguish the positive and negative retrieval instances due to the instance label being unknown. Once the wrong positive instances are introduced, huge noise confuses the model and results in incorrect predictions. How to use the knowledge of the retrieval instances in the inference stage will be researched in the future.

7.3. The implications of RCL

Theoretically, we first verify the feasibility of our method and provide a new paradigm for retrieval enhancement technology. Next, we experimentally verify the superiority of RCL. At the same time, we analyze the performance of RCL on different models and summarize the shortcomings of existing models, which can inspire the counterparts to improve their models and achieve better results. The proposed RCL is a framework that aims to help the existing ALSC model obtain polished instance representations for better sentiment prediction. In practice, RCL is adaptable, flexible, and compatible with the current ALSC models, which can greatly improve the performance of the model without increasing the model parameters, especially for the PLM-based methods. After training with RCL, the model possesses three capabilities at least: (1) understanding the unseen aspects that are not been trained at all; (2) filtering the noise information in the text and only important information remains; (3) being aware of the implicit opinions in the text, which are not expressed with sentiment words.

7.4. The future of RCL

Despite SBERT being a good dense retriever for retrieving similar aspects and instances, the retrieval instances are still unsatisfied as shown in the case study in Section 5.3.1. On the one hand, limited by the scale of the training set, there are not enough high-quality instances for retrieval. In this case, increasing the data size is a feasible research direction in the future. On the other hand, both Glove-based and SBERT-based retrievers are designed for general text calculation tasks, and thus do not perform well to meet our requirements. Hence, fine-tuning the dense retriever to fit our specific demand is another urgent problem to be solved in the future.

CRediT authorship contribution statement

Zhongquan Jian: Conceptualization, Methodology, Conducted experimental verification, Developed the theory and performed the method analysis, Writing – original draft. **Jiajian Li:** Contributed to the experiments, Contributed to the final manuscript. **Qingqiang Wu:** Supervised the findings of the work, Discussed the results and Contributed to the final manuscript. **Junfeng Yao:** Supervised the findings of the work, Discussed the results and Contributed to the final manuscript.

Data availability

Data will be made available on request.

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